



Efficacy and safety of apatinib for patients with advanced extremity desmoid fibromatosis: a retrospective study

Chuanxi Zheng¹ · Jianguo Fang¹ · Yitian Wang¹ · Yong Zhou¹ · Chongqi Tu¹ · Li Min¹ 

Received: 16 October 2020 / Accepted: 5 December 2020 / Published online: 15 January 2021
© Springer-Verlag GmbH Germany, part of Springer Nature 2021

Abstract

Purpose Desmoid fibromatosis (DF) is a locally aggressive connective-tissue tumor arising in deep soft tissues. Although multiple therapeutic modalities have been demonstrated effective for DF, there is no standard systemic treatment for progressive and recurrent DF. As a part of systemic treatment, tyrosine kinase inhibitors have shown promising activity against DF with tolerable toxicity profiles. Thus, the aim of this study was to investigate the efficacy and safety of apatinib, a novel multi-target angiogenesis inhibitor, in patients with DF.

Methods We retrospectively analyzed the medical records of patients with advanced extremity DF regularly treated with apatinib between October 2017 and January 2020 in our center. Apatinib was initially administered with a dose of 250 mg daily and the dose was adjusted according to the toxicity. Tumor response was assessed by the Response Evaluation Criteria in Solid Tumors 1.1 criteria. The primary endpoint was progression-free survival (PFS); objective response rates and drug-related adverse events were also evaluated.

Results A total of 22 (6 male, 16 female) patients with advanced extremity DF were included. The mean medication time was 17 months. None of the patients reached a complete response, but ten (45.5%) patients achieved partial response, and 11 patients (50%) achieved stable disease. One (4.5%) patient developed progressive disease, and the 1-year PFS rate was 95.2%. The disease control rate was 95.4% (21/22) and the objective response rate was 45.5% (10/22). Meanwhile, 18 (81.8%) patients with a tumor shrinkage were accompanied by a decreased signal intensity of lesions in T2-weighted magnetic resonance imaging. The most frequent adverse events included hand-foot syndrome ($n = 7$, 31.8%), fatigue ($n = 6$, 27.2%), local pain ($n = 4$, 18.1%), diarrhea ($n = 4$, 18.1%).

Conclusion Apatinib is an effective and well-tolerated option for patients with advanced extremity DF. Indeed, further prospective, randomized studies with larger cases are required to fully explore the clinical utility of apatinib in DF.

Keywords Desmoid fibromatosis · Apatinib · Tyrosine kinase inhibitor · Efficacy · Safety

Introduction

Desmoid fibromatosis (DF) is a locally aggressive connective-tissue tumor developing in deep soft tissues and has a high tendency to local recurrence but an inability to metastasize (Organization and Cancer 2013). DF is a rare entity with an incidence of 2–4 cases per one million of the

population per year and a peak age of 30–40 years (Penel et al. 2016). Virtually, it can occur in any anatomical location; the primary sites of involvement are the extremities, abdominal wall and abdominal mesentery (Ganeshan et al. 2019). Within these anatomic locations, extremity DF frequently presages a higher risk of recurrence and worse prognosis than abdominal tumors, which pose more difficulties for surgeons (Gronchi et al. 2003).

Although spontaneous regression has been reported in up to 20% of patients with DF, the National Comprehensive Cancer Network guidelines (Version 2.2020) state that observation is just an option for selected patients (small-size tumors, asymptomatic) (Bonvalot et al. 2008; Fiore et al. 2009). For patients with symptomatic or progressive disease, surgical resection is still the mainstay of

Chuanxi Zheng and Jianguo Fang contributed equally to this work.

✉ Li Min
minli1204@scu.edu.cn

¹ Department of Orthopedics, West China Hospital, Sichuan University, Guoxue Xiang No. 37, Chengdu 610041, Sichuan, People's Republic of China

the treatment (Briand et al. 2014; Ganeshan et al. 2019). However, complete surgical excision is not associated with better local tumor control, which in turn causes excessive morbidity (Wang et al. 2015). Local radiotherapy is considered for patients with positive resection margins and those with recurrent or unresectable diseases (Gluck et al. 2011; Kasper et al. 2017a). The combination of surgery and postoperative radiotherapy can reduce local recurrence risk compared with surgical resection alone (Yao et al. 2014). Systemic treatments are usually indicated in patients with recurrent or progressive disease who are not amenable to surgery or radiotherapy. Systemic therapy options include hormonal blockade (tamoxifen combination with sulindac) and cytotoxic chemotherapy (doxorubicin, methotrexate and vinblastine) (Gronchi et al. 2014; Keus et al. 2013). Whereas the response rate associated with antiestrogenic treatments is low and the relationship with therapeutic effectiveness has not yet been fully clarified (Fiore et al. 2015). Meanwhile, the results among different chemotherapy regimens are variable (15–60%) (Elnekave et al. 2018; Palassini et al. 2017; Wang et al. 2019b). Long periods of chemotherapy may increase cumulative dose-dependent cardiotoxicity and potentially impair the fertility of young patients who constitute the dominant population of DF (Kasper et al. 2015; Penel et al. 2016).

Recently, tyrosine kinase inhibitors (TKI) as a new nonchemotherapeutic systemic treatment, such as imatinib, sorafenib, and pazopanib, have exhibited promising clinical results in patients with advanced DF (Chugh et al. 2010; Gounder et al. 2011, 2018; Szucs et al. 2017). Apatinib, a novel oral TKI, highly and selectively targets vascular endothelial growth factor receptor (VEGFR)-2 with a half-maximal inhibitory concentration of 2 nmol/L (Tian et al. 2011). It also synchronously inhibits VEGFR-1, platelet-derived growth factor receptor (PDGFR), and hepatocyte factor receptor (c-KIT) (Tian et al. 2011). Apatinib was approved by the Chinese Food and Drug Administration (CFDA) to treat advanced gastric cancer in 2014. In our off-label experience, this agent possesses effective and durable responses in several advanced soft tissue sarcomas, including synovial sarcoma and alveolar soft part sarcoma (Wang et al. 2020; Wang et al. 2019a). However, the role of apatinib in DF is presently unclear. Thus, the aim of this study was to investigate the efficacy and safety of apatinib in patients with advanced extremity DF.

Patients and methods

Patient enrolling

The medical data of patients with DF treated with apatinib between October 2017 and January 2020 in our center were

retrospectively analyzed. The eligibility criteria were as follows: (1) diagnosis histologically or cytologically confirmed as aggressive DF; (2) tumor located in extremity, which included tumor in the upper or lower extremity, shoulder girdle or buttock; (3) patient had progressive disease (recurrent or primary) that was inoperable for surgery; (4) or patients who refused radiotherapy, chemotherapy, or surgery. Progressive disease was evaluated within a follow-up time of 3 months and the maximum unidimensional measurement of the lesions increased more than 20% on magnetic resonance imaging (MRI). The unresectable lesion was defined as that complete resection of the tumor was infeasible due to the tumor size, tumor location or would cause unacceptable morbidity after extensive surgery. Baseline information consisted of age, gender, presentation status of lesions, tumor location, tumor size, previous treatment history, date and dose of initial apatinib administration, treatment interruption or discontinuation, and progression time. Written informed consent of off-label treatment was obtained from all patients prior to treatment. This study was performed according to the principles embodied in the Declaration of Helsinki and the Institutional Review Board of Sichuan University West China Hospital.

Treatment and response assessment

Apatinib was orally administrated at an initial dose of 250 mg once daily and one treatment cycle was continuous for 28 days. The dose was reduced to 125 mg per day if patient suffered unacceptable or uncontrolled toxicity (grade 3/4 adverse events). If the excessive adverse events could not be adequately remitted by dose reduction and symptomatic management, the treatment was interrupted. When the patient had progressive disease according to Response Evaluation Criteria in Solid Tumors (RECIST) 1.1, the dose was increased to 500 mg once daily (Eisenhauer et al. 2009). If the patient still did not reach disease stabilization or occurred unacceptable side effects in the subsequent two cycles of treatment with 500 mg, then the treatment with apatinib was terminated.

The baseline assessment included radiographic evaluation of the measurable lesions, physical examination, and routine blood tests. Tumor response was defined as a complete response (CR,) partial response (PR), stable disease (SD), or progressive disease (PD) according to RECIST criteria 1.1 (Eisenhauer et al. 2009). Progression-free survival (PFS) was the primary endpoint and identified as the time from the initiation of apatinib administration to the date of disease progression, death, or lost to follow up. Disease control rate (DCR) was defined as the percentage of patients without disease progression on record. The objective response rate (ORR) was defined as the percentage of patients with CR and PR. Drug-related adverse events were graded using the

National Cancer Institute Common Terminology Criteria for Adverse Events (version 4.0).

The patients' clinical response was based on evaluating pain palliation and function of the involved joint. The pain level was subjectively described by patients and quantitated by visual analog scale (VAS) score (Karcioglu et al. 2018). The involved joint range of motion was objectively measured by physical examination. These items were employed prospectively during routine clinical follow-up evaluation and then retrospectively reviewed by clinicians. Finally, quantitative change of tumor signal in T2-weighted MRI was objectively computed using the modified Choi technique described by Stacchiotti et al.: the radiologist drew the largest possible circular region of interest within the greatest cross-sectional diameter of the tumor and the adjacent normal skeletal muscle was selected as another region of interest to provide references. Then, the ratio between the mean tumor signal intensity and the reference muscle signal was obtained (Stacchiotti et al. 2009). These measurements were repeated at the same location of the tumor on subsequent follow-up examinations, and percent changes relative to baseline were calculated.

Statistical analysis

Statistical analysis was performed using SPSS version 22 (IBM Corporation, Armonk, NY, USA) and GraphPad Prism (version 8). Standard descriptive statistics were employed for continuous variables and categorical variables (number and percentage) to characterize the patient demographics, treatment response, and toxicity experience. The last date for the assessment of clinical outcomes was May 30, 2020. PFS was calculated by use of the Kaplan–Meier method. A two-tailed p -values < 0.05 was considered statistically significant.

Results

Patients' characteristics

From October 2017 and January 2020, a total of 22 (6 male, 16 female) patients with advanced extremity DF treated with apatinib were identified. The demographics and clinical baseline characteristics of patients before apatinib treatment were summarized in Table 1. The median age was 26.5 (range 9–53) years. Female patients accounted for 72.7% of the patients. The primary tumor location was thigh in seven patients (31.8%), gluteal region in three patients (13.6%), upper arm in two patients (9.1%), axillary region in two patients (9.1%), scapular region in two patients (9.1%), hand in one patient (4.5%), forearm in one patient (4.5%), leg in one patient (4.5%), foot in one patient (4.5%) and popliteal region in one patient (4.5%).

Table 1 Demographics and clinical baseline characteristics of the patients

Characteristics	Number of patients (percentage and range)
Age	
Median age (year)	26.5
Range	9–53
Sex	
Male	6 (27.2%)
Female	16 (72.8%)
Tumor location	
Thigh	7 (31.8%)
Gluteal region	3 (13.6%)
Upper arm	2 (9.1%)
Axillary region	2 (9.1%)
Scapular region	2 (9.1%)
Hand	2 (9.1%)
Forearm	1 (4.5%)
Leg	1 (4.5%)
Foot	1 (4.5%)
Popliteal region	1 (4.5%)
Unresectable lesions	
Recurrent	14 (63.6%)
Untreated	3 (13.7%)
Resectable lesions	
Recurrent	4 (18.2%)
Untreated	1 (4.5%)
Treatment history	
Surgery	18 (81.8%)
Chemotherapy	4 (18.2%)
Radiotherapy	1 (4.5%)

Most of the enrolled patient had received previous treatment except four patients (18.2%) who did not receive any systemic or surgical therapy prior to apatinib treatment. Three untreated patients (13.7%) and 14 recurrent patients (63.6%) were identified as having unresectable diseases. One untreated patient (4.5%) and four recurrent patients (18.2%) were identified as having resectable diseases, but these patients declined surgery, chemotherapy or radiotherapy. Additionally, combination chemotherapy (doxorubicin and vinblastine based regimen) had been administrated in four patients (18.2%) and one patient (4.5%) received the previous radiotherapy.

Treatment outcomes

The treatment characteristics of apatinib in this study were summarized in Table 2. None of the patients was lost to follow up or die of the disease by the last scheduled visit. The mean medication duration was 17 months (range

Table 2 Clinical features and treatment responses of patients with advanced DF

Patients	Age	Gender	Tumor sites	Treatment history	Initial dose	Medication time (M)	Tumor shrinkage	T2 MRI signal change (%)	RECIST 1.1 response	PFS (M)
1	53	Male	Scapular region	Surgical resection	250 mg	28	42%	− 70	PR	NA
2	33	Female	Thigh	Surgical resection	250 mg	27	53%	− 40	PR	NA
3	18	Female	Popliteal region	Surgery/chemotherapy	250 mg	18	NA	− 10	SD	NA
4	26	Female	Gluteal region	Surgical resection	250 mg	28	82%	− 40	PR	NA
5	24	Female	Thigh	Surgical resection	250 mg	25	50%	− 80	PR	NA
6	18	Female	Scapular region	Surgery/chemotherapy	250 mg	19	7%	− 50	SD	NA
7	20	Female	Thigh	Surgical resection	250 mg	21	60%	− 90	PR	NA
8	50	Female	Hand	Surgery/chemotherapy	250 mg	17	20%	− 60	SD	NA
9	34	Male	Forearm	Surgical resection	250 mg	19	76%	− 80	PR	NA
10	24	Female	Hand	Surgical resection	250 mg	17	19%	− 70	SD	NA
11	21	Female	Thigh	Surgical resection	250 mg	17	25%	− 60	SD	NA
12	29	Female	Upper arm	Untreated	250 mg	16	49%	− 20	PR	NA
13	31	Female	Axillary region	Surgery/chemotherapy	250 mg	15	37%	− 60	PR	NA
14	15	Female	Upper arm	Surgical resection	250 mg	15	NA	− 20	SD	NA
15	21	Male	Thigh	Surgical resection	250 mg	14	21%	− 40	SD	NA
16	23	Female	Leg	Surgical resection	250 mg	14	60%	− 50	PR	NA
17	24	Female	Gluteal region	Surgical resection	250 mg	13	50%	− 70	PR	NA
18	16	Male	Thigh	Untreated	250 mg	12	12%	− 40	SD	NA
19	40	Male	Foot	Surgery/radiotherapy	250 mg	11	NA	5	PD	9
20	23	Female	Gluteal region	Untreated	250 mg	9	3%	− 40	SD	NA
21	38	Female	Axillary region	Untreated	250 mg	8	14%	− 30	SD	NA
22	9	Male	Thigh	Surgical resection	250 mg	11	25%	− 50	SD	NA

PR partial response, SD stable disease, PD progressive disease, PFS progression-free survival, NA not achieved

8~28 months). Although no patient achieved CR, ten patients (45.4%) achieved PR and 11 patients (50%) obtained SD. The overall DCR and ORR was 95.4%, 45.4%, respectively. One patient (4.5%) developed PD after 9 months of apatinib treatment and the 1-year PFS rates were 95.2%. However, the median PFS was not reached at the time of analysis. This patient was re-recurrent DF following two surgeries and he temporarily obtained disease stabilization lasting 9 months with apatinib treatment. After the disease progression, he received additional two cycles of treatment with apatinib dose of 500 mg. Ultimately, this treatment did not provide the potential to control disease progression; he chose amputation surgery due to the significant influence of symptoms on daily activities.

With apatinib treatment, the reduction of tumor size was observed in 19 patients (86.3%). The mean tumor shrinkage in all patients with PR was 58.9% (95% CI 46.1–71.6%), and nine patients with SD obtained tumor shrinkage with a mean tumor reduction of 16.2% (95% CI 10.2–22.1%) (Fig. 1). Intriguingly, we noticed that signal intensity decreases of lesions in T2-weighted MRI occurred in all patients with tumor shrinkage. In patients with PR and SD, the mean relative decrease of the signal intensity was 61.0%, 42.7%,

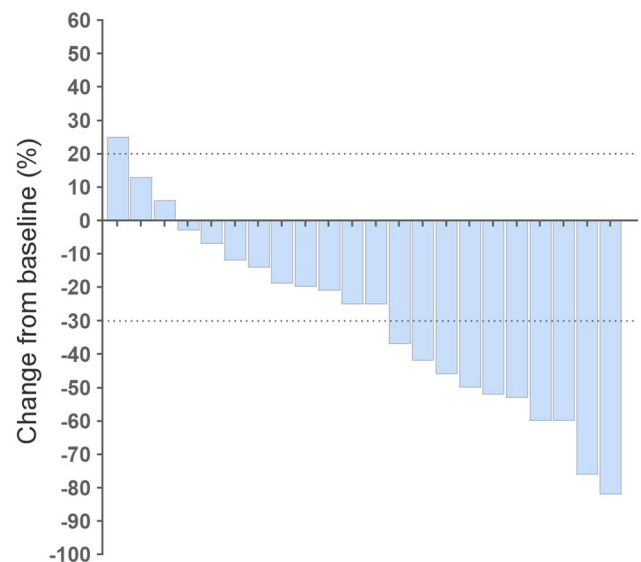


Fig. 1 Waterfall plot of tumor shrinkage from baseline for 22 patients with advanced extremity DF

respectively. There was no statistical difference between the signal value decrease of patients with PR and SD

($p=0.0544$). A 20-year patient with progressive DF and 21 months of treatment course presents a typical therapeutic efficacy, shown in Fig. 2. In addition, all patients with tumor reduction synchronously reported symptomatic improvement after treatment in terms of pain relief and function of the involved joint. The mean pretreated VAS score of patients with tumor reduction was 4.7 and the mean post-treated VAS score was 2. A significant improvement was observed in the VAS score following treatment. The improvement of the joint mobility in patient with tumor reduction following treatment was summarized in Table 3.

Safety and toxicity

The mean medication duration was 17 months (95% CI 14.4–19.6). All patients were administrated with an initial apatinib dose of 250 mg daily, but two patients (9.1%) had to reduce the dose due to intolerable drug-related toxicity. None of patient discontinued treatment because of uncontrolled adverse events. The common adverse events included hand-foot skin syndrome ($n=7$, 31.8%), fatigue ($n=6$, 27.2%), local pain ($n=4$, 18.1%), diarrhea ($n=4$, 18.1%)

and hair hypopigmentation ($n=4$, 18.1%). The majority of adverse events were grades 1 or 2 and well-controlled by the symptomatic treatment. The grade-3 adverse events comprised hand-foot skin syndrome ($n=2$, 9.1%), which were relieved after dose reduction. No grade-4 adverse events or lethal drug-related toxicity were reported. The drug-related adverse events that occurred during the treatment were summarized in Table 4.

Discussion

As a new systemic treatment, anti-angiogenesis has contributed a crucial part to DF management, particularly for patients with advanced disease. TKIs block phosphorylation of the ligand-activated receptor and activate mitogen-activated kinase, ultimately inhibiting tumor angiogenesis and cellular proliferation (Yang et al. 2020). Imatinib was the first TKI used for management of aggressive DF, which exerted an inhibitory effect on multiple class 3 receptor tyrosine kinases, including PDGFR, PDGFR- β , and c-KIT (Skubitz 2017). In this context, inhibition of PDGFR- β kinase associated with the critical pathogenetic

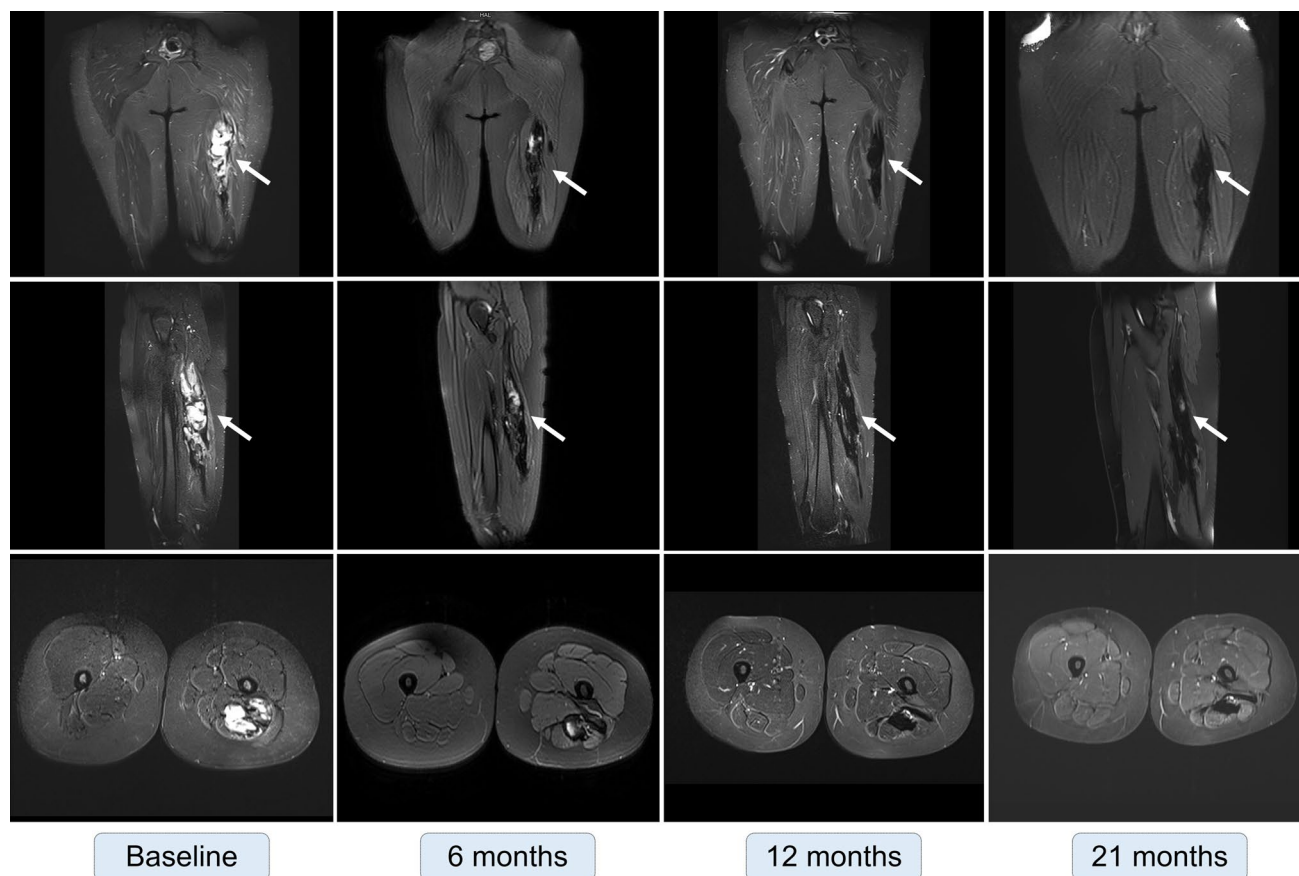


Fig. 2 A 20-year patient with advanced extremity DF reached partial disease after 21 months of apatinib treatment

Table 3 Clinical response of patients with advanced DF treated with apatinib

Patients	Tumor site	Involved joint	RECIST 1.1 Response	Pre/post-treated pain level (VAS)	Pre/post-treated range of motion (°)					Adduction	External rotation	Internal rotation
					Flexion	Extension	Abduction	Abduction	Abduction			
1	Scapular region	Shoulder	PR	4/1	90/90	20/30	70/90	40/40	10/30	50/70		
2	Thigh	Hip	PR	6/3	90/120	10/20	30/40	30/30	40/40	30/40		
3	Popliteal region	Knee	SD	6/4	70/100	5/0	NA	NA	NA	NA		
4	Gluteal region	Hip	PR	4/1	90/120	10/20	20/50	20/30	30/40	40/40		
5	Thigh	Hip	PR	4/1	80/110	10/10	40/50	30/30	30/40	40/40		
6	Scapular region	Shoulder	SD	6/3	90/90	30/30	80/90	40/40	10/20	60/70		
7	Thigh	Hip	PR	5/1	70/100	10/20	30/40	20/30	40/40	30/40		
8	Hand	MCP joint	PR	6/3	50/70	5/20	NA	NA	NA	NA		
9	Forearm	Wrist joint	PR	5/1	20/30	10/20	NA	NA	20/60 ^a	30/80 ^b		
10	Hand	MCP joint	SD	5/3	60/80	10/10	NA	NA	NA	NA		
11	Thigh	Hip	SD	3/0	90/100	0/10	30/30	20/20	40/40	40/40		
12	Upper arm	Elbow	PR	6/2	110/130	30/0	NA	NA	NA	NA		
13	Axillary region	Shoulder	SD	3/0	60/90	30/50	40/60	10/40	40/60	50/70		
14	Upper arm	Elbow	SD	4/3	90/90	40/20	NA	NA	NA	NA		
15	Thigh	Hip	SD	4/2	100/130	0/15	30/30	20/40	40/40	40/40		
16	Leg	Knee	PR	3/1	70/120	20/0	NA	NA	NA	NA		
17	Gluteal region	Hip	PR	6/2	100/130	5/10	10/30	10/20	40/40	40/40		
18	Thigh	Hip	SD	5/3	70/120	0/10	40/40	30/30	40/40	30/30		
19	Foot	Ankle	PD	6/5	20/NA	30/NA	NA	NA	NA	NA		
20	Gluteal region	Hip	PR	5/3	70/120	0/15	20/30	10/20	20/40	10/40		
21	Axillary region	Shoulder	SD	4/3	70/90	10/40	50/70	30/40	20/40	50/70		
22	Thigh	Hip	SD	4/1	80/120	0/15	20/50	10/30	20/40	20/40		

NA not achieved, MCP metacarpophalangeal;

^aRange of pronation^bRange of supination

Table 4 Adverse events of apatinib treatment

Adverse events	Total, n (%)	Grade 1	Grade 2	Grade 3–4
Hand-foot syndrome	7 (31.8%)	3	2	2
Fatigue	6 (27.2%)	6	0	0
Local pain	4 (18.1%)	2	2	0
Diarrhea	4 (18.1%)	4	0	0
Hair hypopigmentation	4 (18.1%)	3	1	0
Oral mucositis	3 (13.6%)	3	0	0
Vomiting	3 (13.6%)	3	0	0
Hypertension	3 (13.6%)	1	2	0
Insomnia	3 (13.6%)	2	1	0
Skin hypopigmentation	2 (9.1%)	2	0	0
Abdominal distention	2 (9.1%)	2	0	0
Rash acneiform	2 (9.1%)	1	1	0
Proteinuria	2 (9.1%)	1	1	0
Anemia	1 (4.5%)	1	0	0
Thrombocytopenia	1 (4.5%)	1	0	0
Elevated aminotransferase	1 (4.5%)	1	0	0
Anorexia	1 (4.5%)	1	0	0
Nausea	1 (4.5%)	1	0	0

Wnt/ β -catenin pathway was considered as the molecular basis for tumor response to imatinib (Heinrich et al. 2006). Generally, imatinib has exhibited a comparable DCR of 60–80% despite a relatively low ORR of 6–19%, with a mild to moderate toxicity profile (Chugh et al. 2010; Heinrich et al. 2006; Kasper et al. 2017b; Penel et al. 2011). Szucs et al. reported that three patients achieved PR and the remaining five patients obtained SD after pazopanib treatment, with a median PFS of 13.5 months (Szucs et al. 2017). The further results of phase II randomized study showed that 37% of patients had PR and 58.7% of patients had SD in pazopanib treatment group, with a 1-year PFS survival rate of 85.6% (Toulmonde et al. 2019). In a retrospective non-randomized study, Gounder et al. demonstrated that sorafenib treatment induced PR in 25% of patients and 70% of patients reached disease stabilization (Gounder et al. 2011). Subsequently, a randomized phase III clinical trial of sorafenib demonstrated that patients with advanced DF obtained an ORR of 33% with the use of sorafenib, but 20% of patients in the placebo arm also had spontaneous disease regression (Gounder et al. 2018). Notably, a smaller percentage of patients in the sorafenib group presented disease progression rather than placebo group (12% vs 63%), with the 1-year and 2-year PFS of 89% and 81%, respectively (Gounder et al. 2018). In the present study, the 1-year PFS rate of patients was 95.2%, similar to that of sorafenib and pazopanib (Gounder et al. 2018; Toulmonde et al. 2019). This improvement in PFS induced by both sorafenib and apatinib may be correlated

with multi-target nature of these tyrosine kinase inhibitors for VEGFR, PDGFR- β , while further studies are required to confirm this speculation (Tian et al. 2011; Wilhelm et al. 2004).

As DF is characterized by locally infiltrative growth without metastatic potential, the primary therapeutic purpose is to slow or halt disease progression and improve life quality. Therefore, we chose an initial dose of apatinib (250 mg daily) which was lower than the dose routinely used in soft tissue sarcomas to balance the efficacy of drug use against toxic side effects. In terms of safety, the most common toxicity of apatinib observed in the present study included hand-foot syndrome, fatigue, diarrhea, and local pain, similar to those reported in previous literature of apatinib in patients with advanced metastatic soft-tissue sarcoma (Liao et al. 2019; Liu et al. 2020). Importantly, no grade-4 adverse events were observed in the present study, although two patients (9.1%) suffered grade-3 adverse events of hand-foot skin syndrome. Ultimately, no dose interruption or discontinuing of apatinib was reported in the current treatment. However, up to 45% of patients interrupted or discontinued sorafenib or imatinib administration due to the unacceptable drug-related toxicities (Luo et al. 2019; Penel et al. 2011). Moreover, 12.5% of patients interrupted pazopanib treatment because of uncontrollable grade-3 hypertension (Szucs et al. 2017). The overall toxicity of apatinib was relatively well tolerated, while hand-foot syndrome, gastrointestinal discomfort (such as diarrhea, vomiting, nausea), local pain should be noticed.

Currently, the sensitivity of using RECIST-defined responses to assess treatment effect and survival among patients with solid tumors is controversial. Sheth et al. reported that there was discordance among different response assessments of DF patients based on unidimensional measurement (RECIST), tumor volume, and T2-weighted signal intensity (Sheth et al. 2016). Teixeira et al. showed that the low T2-weighted signal intensity indicated a general tendency for stability or regression. In contrast, the high T2-weighted signal intensity denoted a progressive tendency, suggesting that change of T2-weighted signal intensity was correlated with the biological response of this tumor (Gondim Teixeira et al. 2020). Therefore, the current RECIST metric may underestimate treatment efficacy. In the present study, signal intensity decreases of lesions in T2-weighted MRI occurred in all patients with tumor shrinkage, including nine patients without a distinct tumor size decrease (7~25%). These patients also reported pain relief and improved function of adjacent joint that might enhance the patients' health-related quality of life as the patients with PR. Results from the present study suggest that MRI T2-weighted signal intensity and physical examination findings may be the additional items that can potentially be used to evaluate treatment efficacy on DF.

In addition, our study has several limitations. This is a retrospective review of one institution, and there was no control for comparison. Meanwhile, we only enrolled 22 patients with a mean medication duration of 17 months, and the optimal duration of apatinib treatment for patients with advanced DF remains unknown. Finally, the definite mechanism of apatinib against DF was not clear. The present study results were limited to clinical evaluations, lacking the relevant molecular biology assessment in each patient, such as catenin beta-1 (CTNNB1), PDGFR- β , c-KIT, or the adenomatous polyposis coli (APC) mutation status that related to the activation of Wnt/ β -catenin pathway.

Conclusion

Our study provides the first clinical evidence of the efficacy and safety of apatinib in patients with DF. Apatinib showed a notable activity and well-tolerated profile in patients with advanced extremity DF. Although our study results may be preliminary, treatment with apatinib indeed slows the disease progression of patients with DF. Further prospective, randomized studies with larger cases are required to fully explore the clinical utility of apatinib in DF and hopefully provide an alternative approach to manage this refractory disease.

Funding This work was supported, in part, by the Chengdu Science and Technology Program Projects (2017-CY02-00032-GX), National Natural Science Foundation of China (No. 81702664 and 81801852), and National Key Research and Development Program of China (No. 2016YFC1102003 and 2017YFB0702604).

Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Compliance with ethical standards

Conflict of interest The authors report no conflicts of interest in this work.

Ethical approval This retrospective study was performed using data from anonymized patients who received apatinib treatment between October 2017 and January 2020. Because of the nature of retrospective design and patient anonymization, the ethical board of Sichuan University West China Hospital approved the retrospective study.

Informed consent Written informed consent of off-label treatment was obtained from all patients before treatment.

References

Bonvalot S et al (2008) Extra-abdominal primary fibromatosis: aggressive management could be avoided in a subgroup of patients European journal of surgical oncology : the journal

of the European Society of Surgical Oncology and the British Association of. *Surg Oncol* 34:462–468. <https://doi.org/10.1016/j.ejso.2007.06.006>

Briand S, Barbier O, Biau D, Bertrand-Vasseur A, Larousserie F, Anract P, Gouin F (2014) Wait-and-see policy as a first-line management for extra-abdominal desmoid tumors. *J Bone Jt Surg Am* 96:631–638. <https://doi.org/10.2106/jbjs.M.00988>

Chugh R et al (2010a) Efficacy of imatinib in aggressive fibromatosis: results of a phase II multicenter Sarcoma Alliance for Research through Collaboration (SARC) trial. *Clin Cancer Res* 16:4884–4891. <https://doi.org/10.1158/1078-0432.Ccr-10-1177>

Eisenhauer EA et al (2009) New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer* 45:228–247. <https://doi.org/10.1016/j.ejca.2008.10.026>

Elnekave E et al (2018) Doxorubicin-eluting intra-arterial therapy for pediatric extra-abdominal desmoid fibromatoses: a promising approach for a perplexing disease. *J Vasc Interv Radiol* 29:1376–1382. <https://doi.org/10.1016/j.jvir.2018.04.009>

Fiore M et al (2009) Desmoid-type fibromatosis: a front-line conservative approach to select patients for surgical treatment. *Ann Surg Oncol* 16:2587–2593. <https://doi.org/10.1245/s10434-009-0586-2>

Fiore M et al (2015) Hormonal manipulation with toremifene in sporadic desmoid-type fibromatosis. *Eur J Cancer* 51:2800–2807. <https://doi.org/10.1016/j.ejca.2015.08.026>

Ganesan D, Amini B, Nikolaidis P, Assing M, Vikram R (2019) Current update on desmoid fibromatosis. *J Comput Assist Tomogr* 43:29–38. <https://doi.org/10.1097/RCT.0000000000000790>

Gluck I, Griffith KA, Biermann JS, Feng FY, Lucas DR, Ben-Josef E (2011) Role of radiotherapy in the management of desmoid tumors. *Int J Radiat Oncol Biol Phys* 80:787–792. <https://doi.org/10.1016/j.ijrobp.2010.02.053>

Gondim Teixeira PA, Biouchi H, Abou Arab W, Rios M, Sirveaux F, Hossu G, Blum A (2020) Evidence-based MR imaging follow-up strategy for desmoid-type fibromatosis. *Eur Radiol* 30:895–902. <https://doi.org/10.1007/s00330-019-06404-4>

Gounder MM et al (2011) Activity of Sorafenib against desmoid tumor/deep fibromatosis. *Clin Cancer Res* 17:4082–4090. <https://doi.org/10.1158/1078-0432.Ccr-10-3322>

Gounder MM et al (2018) Sorafenib for advanced and refractory desmoid tumors. *N Engl J Med* 379:2417–2428. <https://doi.org/10.1056/NEJMoa1805052>

Gronchi A et al (2003) Quality of surgery and outcome in extra-abdominal aggressive fibromatosis: a series of patients surgically treated at a single institution. *J Clin Oncol* 21:1390–1397. <https://doi.org/10.1200/JCO.2003.05.150>

Gronchi A et al (2014) Sporadic desmoid-type fibromatosis: a stepwise approach to a non-metastasising neoplasm—a position paper from the Italian and the French Sarcoma Group. *Ann Oncol* 25:578–583. <https://doi.org/10.1093/annonc/mdt485>

Heinrich MC et al (2006) Clinical and molecular studies of the effect of imatinib on advanced aggressive fibromatosis (desmoid tumor). *J Clin Oncol* 24:1195–1203. <https://doi.org/10.1200/jco.2005.04.0717>

Karcioglu O, Topacoglu H, Dikme O, Dikme O (2018) A systematic review of the pain scales in adults: Which to use? *Am J Emerg Med* 36:707–714. <https://doi.org/10.1016/j.ajem.2018.01.008>

Kasper B et al (2015) Management of sporadic desmoid-type fibromatosis: a European consensus approach based on patients' and professionals' expertise—a sarcoma patients EuroNet and European Organisation for Research and Treatment of Cancer/Soft Tissue and Bone Sarcoma Group initiative. *Eur J Cancer* 51:127–136. <https://doi.org/10.1016/j.ejca.2014.11.005>

Kasper B et al (2017a) An update on the management of sporadic desmoid-type fibromatosis: a European Consensus Initiative between Sarcoma Patients EuroNet (SPAEN) and European Organization for Research and Treatment of Cancer (EORTC)/

- Soft Tissue and Bone Sarcoma Group (STBSG). *Ann Oncol* 28:2399–2408. <https://doi.org/10.1093/annonc/mdx323>
- Kasper B et al (2017b) Imatinib induces sustained progression arrest in RECIST progressive desmoid tumours: final results of a phase II study of the German Interdisciplinary Sarcoma Group (GISG). *Eur J Cancer* 76:60–67. <https://doi.org/10.1016/j.ejca.2017.02.001>
- Keus RB et al (2013) Results of a phase II pilot study of moderate dose radiotherapy for inoperable desmoid-type fibromatosis—an EORTC STBSG and ROG study (EORTC 62991–22998). *Ann Oncol* 24:2672–2676. <https://doi.org/10.1093/annonc/mdt254>
- Liao Z et al (2019) Phase II trial of VEGFR2 inhibitor apatinib for metastatic sarcoma: focus on efficacy and safety. *Exp Mol Med* 51:1–11. <https://doi.org/10.1038/s12276-019-0221-7>
- Liu X et al (2020) Efficacy and safety of the VEGFR2 inhibitor Apatinib for metastatic soft tissue sarcoma: Chinese cohort data from NCT03121846. *Biomed Pharmacother* 122:109587. <https://doi.org/10.1016/j.biopha.2019.109587>
- Luo J et al (2019) Single institution experience of split course radiotherapy in patients with desmoid tumors. *Onco Targets Ther* 12:1741–1748. <https://doi.org/10.2147/OTT.S189449>
- Organization WH, Cancer IAfRo (2013) WHO classification of tumours of soft tissue and bone.
- Palassini E et al (2017) Long-term efficacy of methotrexate plus vinblastine/vinorelbine in a large series of patients affected by desmoid-type fibromatosis. *Cancer journal (Sudbury, Mass)* 23:86–91. <https://doi.org/10.1097/ppo.0000000000000254>
- Penel N et al (2011) Imatinib for progressive and recurrent aggressive fibromatosis (desmoid tumors): an FNCLCC/French Sarcoma Group phase II trial with a long-term follow-up. *Ann Oncol* 22:452–457. <https://doi.org/10.1093/annonc/mdq341>
- Penel N et al (2016) Management of desmoid tumours: a nationwide survey of labelled reference centre networks in France. *Eur J Cancer* 58:90–96. <https://doi.org/10.1016/j.ejca.2016.02.008>
- Sheth PJ et al (2016) Desmoid fibromatosis: MRI features of response to systemic therapy. *Skeletal Radiol* 45:1365–1373. <https://doi.org/10.1007/s00256-016-2439-y>
- Skubitz KM (2017) Biology and treatment of aggressive fibromatosis or desmoid tumor. *Mayo Clin Proc* 92:947–964. <https://doi.org/10.1016/j.mayocp.2017.02.012>
- Stacchiotti S et al (2009) High-grade soft-tissue sarcomas: tumor response assessment—pilot study to assess the correlation between radiologic and pathologic response by using RECIST and Choi criteria. *Radiology* 251:447–456. <https://doi.org/10.1148/radiol.12512081403>
- Szucs Z et al (2017) Pazopanib, a promising option for the treatment of aggressive fibromatosis. *Anticancer Drugs* 28:421–426. <https://doi.org/10.1097/CAD.0000000000000474>
- Tian S et al (2011) YN968D1 is a novel and selective inhibitor of vascular endothelial growth factor receptor-2 tyrosine kinase with potent activity in vitro and in vivo. *Cancer Sci* 102:1374–1380. <https://doi.org/10.1111/j.1349-7006.2011.01939.x>
- Toulmonde M et al (2019) Pazopanib or methotrexate-vinblastine combination chemotherapy in adult patients with progressive desmoid tumours (DESMOPAZ): a non-comparative, randomised, open-label, multicentre, phase 2 study. *Lancet Oncol* 20:1263–1272. [https://doi.org/10.1016/s1470-2045\(19\)30276-1](https://doi.org/10.1016/s1470-2045(19)30276-1)
- Wang YF, Guo W, Sun KK, Yang RL, Tang XD, Ji T, Tang S (2015) Postoperative recurrence of desmoid tumors: clinical and pathological perspectives. *World J Surg Oncol* 13:26. <https://doi.org/10.1186/s12957-015-0450-8>
- Wang Y et al (2019a) The efficacy and safety of apatinib in metastatic alveolar soft part sarcoma: a case series of six patients in one institution. *Cancer Manag Res* 11:3583–3591. <https://doi.org/10.2147/CMAR.S198429>
- Wang Z, Wu J, Tian X, Hao C (2019b) Targeted therapy of desmoid-type fibromatosis: mechanism, current situation, and future prospects. *Front Med* 13:427–437. <https://doi.org/10.1007/s11684-018-0672-6>
- Wang Y et al (2020) The efficacy and safety of apatinib in advanced synovial sarcoma: a case series of twenty-one patients in one single institution. *Cancer Manag Res* 12:5255–5264. <https://doi.org/10.2147/CMAR.S254296>
- Wilhelm SM et al (2004) BAY 43–9006 exhibits broad spectrum oral antitumor activity and targets the RAF/MEK/ERK pathway and receptor tyrosine kinases involved in tumor progression and angiogenesis. *Cancer Res* 64:7099–7109. <https://doi.org/10.1158/0008-5472.Can-04-1443>
- Yang QK, Chen T, Wang SQ, Zhang XJ, Yao ZX (2020) Apatinib as targeted therapy for advanced bone and soft tissue sarcoma: a dilemma of reversing multidrug resistance while suffering drug resistance itself. *Angiogenesis* 23:279–298. <https://doi.org/10.1007/s10456-020-09716-y>
- Yao X, Corbett T, Gupta AA, Kandel RA, Verma S, Werier J, Ghert M (2014) A systematic review of active treatment options in patients with desmoid tumours. *Curr Oncol* 21:e613–629. <https://doi.org/10.3747/co.21.1995>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.